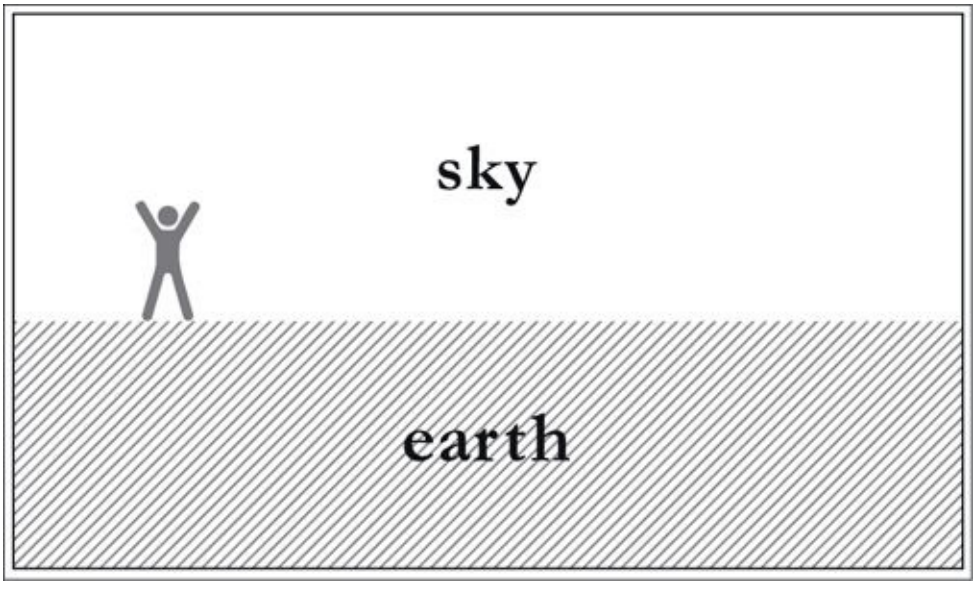
THIRD LESSON

The Architecture of the Cosmos

In the first half of the twentieth century Einstein described the workings of space and time, whilst Niels Bohr and his young disciples captured in equations the strange quantum nature of matter. In the second half of the century physicists built upon these foundations, applying the two new theories to widely varying domains of nature: from the macrocosmic structure of the universe to the microcosm of elementary particles. I speak of the first of these in this lesson, and of the second in the next.

This lesson is made up mostly of simple drawings. The reason for this is that before experiments, measurements, mathematics and rigorous deductions, science is above all about visions. Science begins with a vision. Scientific thought is fed by the capacity to ‘see’ things differently than they have previously been seen. I want to offer here a brief, modest outline of a journey between visions.



This represents how the cosmos was conceptualized for millennia: the Earth below, the sky above. The first great scientific revolution, accomplished by Anaximander twenty-six centuries ago when trying to figure out how it is